

MANAN AGARWAL

☎ +91 9031700201 | ✉ agarwalmanan73@gmail.com | in LinkedIn | GitHub | </> LeetCode

Education

Siddaganga Institute of Technology

Bachelor of Engineering in Computer Science and Engineering | CGPA: 9.46

Sep 2023 – Present

Tumkur, Karnataka

Senior Secondary (CBSE) | **Percentage: 89.2%**

Jun 2023 | Jamshedpur, Jharkhand

Secondary (ICSE) | **Percentage: 95.0%**

Jun 2021 | Jamshedpur, Jharkhand

Relevant Coursework

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (C++), System Design, Database Management Systems, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Cloud Computing, Software Project Management, Compiler Design

Technical Skills

Languages: C++, Python, C

DevOps & Tooling: Git, Docker, Kubernetes, Linux, Postman

Distributed Systems & Microservices: Django, FastAPI, Node.js, Express.js, REST APIs, JWT

Frontend & Web Exposure: React.js, Tailwind CSS, HTML, JavaScript

Database & Streaming: MySQL, MongoDB, Apache Kafka

Projects

CloudSpend AI: AI-Powered FinOps Cost Optimization Platform



- Built an AI-powered cloud cost analysis tool using React, FastAPI, MySQL, and Docker.
- Developed a RAG-based AI assistant using Gemini API, LangChain, and FAISS for cost optimization.
- Designed REST APIs for bill upload, analysis, and temporary data storage.
- Tech Stack:** Python, React, FastAPI, MongoDB, Gemini API, LangChain, FAISS, Docker

SystemCraft: Architect with Clarity



- Built a system design visualizer using **ReactFlow** and **Framer Motion** for interactive drag-and-drop architecture mapping.
- Implemented **JWT authentication** and **REST APIs** for secure user management and persistent design storage.
- Engineered a scalable **MongoDB** backend and optimized React state management for efficient large-scale diagram rendering.
- Tech Stack:** React, ReactFlow, Framer Motion, Node.js, Express.js, JWT, MongoDB

CPU Scheduling And Process Simulator



- Built a multi-threaded **C++ simulator** implementing FCFS, SJF, and Round Robin scheduling.
- Used **POSIX Threads and Mutex Locks** for thread-safe concurrent execution.
- Applied **OOP and inheritance** to create a modular scheduling framework.
- Tech Stack:** C++, POSIX Threads, Mutex Locks, Object-Oriented Programming (OOP)

City Traffic Hotspot Finder

Live

- Engineered a Python **ETL pipeline** to process **1M+ GeoLife GPS records** into Google BigQuery.
- Developed optimized **SQL queries** for efficient spatial-temporal traffic analysis.
- Leveraged **GCP and BigQuery** to identify traffic hotspots and mobility patterns.
- Tech Stack:** Python, SQL, Google BigQuery, Google Cloud Platform (GCP)

Achievements

Secured **1st** position at robotics workshop among **30** teams.

Completed **350+** questions on Leetcode

Completed C++ and DBMS course on Codechef.

Positions Of Responsibility

Decoders - The Official Programming Club of S.I.T. | Member

May 2024 – Present | Tumkur, Karnataka

- Collaborated in a **team** to conduct various **coding** challenges and events, effectively promoting a coding culture on the institute campus.

Avalanche - Feel Accelerated. | Member (Treasurer)

May 2024 – Present | Tumkur, Karnataka

- Coordinated technical workshops and cultural events while collaborating with **Goonj India**, a non-profit dedicated to bridging the urban-rural gap.